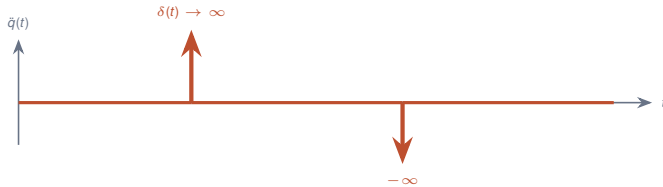
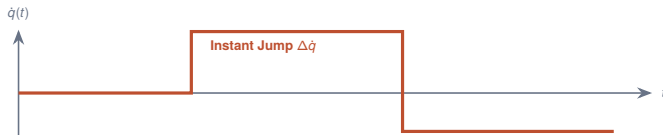


DERIVATIVE CONTINUITY C^2 VS. DISCONTINUOUS WAYPOINT STEPPING

Actuator Protection & Gearbox Shock Mitigation

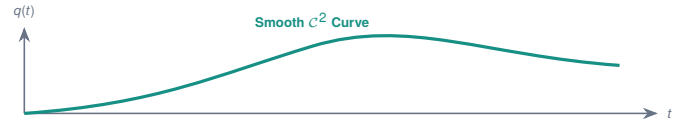
❌ Raw Chunk Stepping (C^0 Failure → Hardware Shock)



⚠️ Hardware Destruction:

- Infinite jerk ($\ddot{\dot{q}} \rightarrow \infty$) creates high-frequency torque shock.
- Shears cycloidal drive pins & harmonic flexspline teeth.
- Excites mechanical chatter; triggers motor overcurrent trip.

✅ C^2 Quintic Spline ($\|\ddot{\dot{q}}\| \leq j_{\max}$ Continuous)



🛡️ Actuator Protection:

- Jerk bounded strictly within $\|\ddot{\dot{q}}\| \leq j_{\max} = 25 \text{ rad/s}^3$.
- Zero gear backlash impact; smooth torque transmission.
- Motor coil current $I(t)$ remains within continuous thermal envelope.